Introduction to Quantitative Methods for Economics and Finance

Problem Set 1

Professor: Alexis Montecinos
Due Date: February 13, at 11:59 pm
Instructions. Show all steps clearly. Interpret numerical results in words where appropriate.

Part A — Expected Value and Standard Deviation

Problem 1: Expected Value

Question. A random variable X takes values $\{1, 2, 4\}$ with probabilities $\{0.2, 0.5, 0.3\}$:

$$P(X = 1) = 0.2, \quad P(X = 2) = 0.5, \quad P(X = 4) = 0.3.$$

(a) Compute the expected value $\mathbb{E}[X]$.

(b) Explain what the expected value represents in this context.

Problem 2: Variance and Standard Deviation

Question. Using the random variable X from Problem 1:

(a) Compute $\mathbb{E}[X^2]$.

(b) Compute the variance $\text{Var}(X)$.

(c) Compute the standard deviation σ_X.

(d) Interpret the meaning of the standard deviation.

Part B — Bernoulli Distribution

Problem 3: Bernoulli Random Variable

Question. A trader receives \$1 if a stock goes up tomorrow and \$0 otherwise. The probability the stock goes up is $p = 0.6$.

(a) Define the random variable X.

(b) Identify the distribution of X.

(c) Compute $\mathbb{E}[X]$, $\text{Var}(X)$, and σ_X.

(d) Explain the economic meaning of $\mathbb{E}[X]$.

Part C — Binomial Distribution

Problem 4: Binomial Outcomes

Question. A salesperson makes $n = 5$ independent sales calls, each with probability of success $p = 0.3$.

(a) Define the random variable X.

(b) Identify the distribution of X.

(c) Compute $P(X = 3)$.

(d) Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Problem 5: Risk Interpretation

Question. Using the binomial model from Problem 4:

(a) What does the standard deviation tell you about the number of successful calls?

(b) Would you expect most outcomes to be close to the expected value? Why?

Part D — Poisson Distribution

Problem 6: Event Counts

Question. The number of customer complaints arriving per hour follows a Poisson distribution with mean $\lambda = 2$ complaints/hour.

(a) Write the probability mass function.

(b) Compute $P(X = 3)$.

(c) Compute $\mathbb{E}[X]$, $\mathrm{Var}(X)$, and σ_X.

(d) Explain why the Poisson distribution is appropriate here.

Problem 7: Comparison Question

Question. Compare the variance and expected value of a Poisson distribution.

(a) What special property do you observe?

(b) How does this differ from the binomial distribution?

Part E — Conceptual Questions

Problem 8: Concept Check

Question. Answer briefly:

(a) When should a Bernoulli distribution be used instead of a binomial?

(b) Why does the Poisson distribution not require a fixed number of trials?

(c) In practical terms, what does a high standard deviation imply?